

R41F_LIVE_VNV_SUMMARY

R41-F — Consolidated Live Validation+Verification Evidence Summary

Revision: 1 **Created:** 2026-07-11 **Last modified:** 2026-07-11 **Status:** active **Track:** (T1/feature/helixllm-full-extension - claude3) **Scope:** §11.4.83 (docs/qa/ end-user evidence mandate) — one reader-friendly index over the R41-F fleet's fresh LIVE evidence, produced 2026-07-11 after the coder was booted rootless (§11.4.111 fix). This document is a **read-only compilation**: nothing it cites was re-run, re-verified, or modified to produce this index — every cell below is drawn directly from the named RESULTS.md (or equivalent) already committed by its owning stream.

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1. Scope and method

On 2026-07-11 the HelixLLM coder (Qwen3-Coder-30B-A3B-Instruct-Q4_K_M.gguf, llama.cpp OpenAI-compatible server, port : 18434) was brought up **rootless** after a CDI (Container Device Interface) stale-binding blocker was root-caused and fixed (§11.4.111 — see §4). With the coder live, four coder-independent-but-coder-exercising R41-F streams produced fresh, committed, real-evidence RESULTS.md documents. This summary:

- Verifies each cited RESULTS.md (or evidence file) **exists on disk** and **skims its verdict/metrics** before citing it (§11.4.6 — no fabricated metric appears below).
- Cites the **exact evidence path + commit hash** for every claim so a reader can open the primary source directly.

- Marks anything not yet landed as **PENDING**, never invents a result (§11.4.6 / §11.4.123).
- Was produced **read-only**: no submodule, /mnt/track1, adb, secret-scan script, git-hook script, or helix_code/internal/server file was touched by this task; the only new file is this document itself.

2. Evidence index

Capability	Live-proven?	Key metric	Evidence path (+ c
Coder boot (rootless)	YES	Fresh nonce HELIX-1783774679-17765 echoed verbatim by the live coder (completion_tokens=20)	docs/qa/coder_boot_live RESULTS.md (+res at time of writing same batch as the l the script; the livep committed)
Dual-wire facade — full HTTP E2E	YES	401 fail-closed (no auth) + OpenAI /v1/chat/completions 200 nonce-echo + Anthropic /v1/messages 200 nonce-echo + live tool-call wire-shape divergence (both shapes), all via real curl+httptest+router+middleware+coder	docs/qa/dualwire_live_e RESULTS.md — cor
Lane-B multi-instance serving (co-resident)	YES	163.67 tok/s mean / 163.34 tok/s median single-stream (matches prior non-co-resident baseline); 3/3 concurrent succeeded; coder llama-server PID identical (1980342) pre/post — never restarted	docs/qa/lane_b_liveproo RESULTS.md (+ben benchmark_scrip
HelixQA coder-bench + concurrency banks	YES	13/13 PASS, 0 FAIL — p50/p95/p99 latency 152/171/187 ms @concurrency-1 scaling to 1076/2113/2128 ms @concurrency-100; throughput 269.7–624.2 tok/s; TTFT p50=3 ms; 50/50 concurrent requests HTTP 200, zero drops, zero duplicate responses	docs/qa/helixqa_live_vn RESULTS.md — cor

Capability	Live-proven?	Key metric	Evidence path (+ c
Generative (image/video) live-proof	PENDING	Not yet on disk at time of writing (docs/qa/generative_liveproof_* absent; find docs/qa -maxdepth 1 -iname "*generative*" returns nothing)	N/A — tracked in the (. superpowers/so as stream ad9a782 a GPU-freed backfill status “BACKFILL . writing, not yet ret

3. Per-capability detail

3.1 Coder boot (rootless)

docs/qa/coder_boot_liveproof_20260711T125759Z/RESULTS.md (2 lines) records:

```
nonce=HELIX-1783774679-17765
PASS: nonce echoed
```

The accompanying response.json in the same directory shows the full raw OpenAI-compatible chat-completion response from /models/Qwen3-Coder-30B-A3B-Instruct-Q4_K_M.gguf echoing that exact nonce ("content": "HELIX-1783774679-17765"), with real timing fields (prompt_ms, predicted_ms, predicted_per_second≈285) — a genuine, non-cached model response, not a fixture. This directory is **currently untracked** (git status shows ??); it was produced by the coder-boot stream in the same session and is expected to land in the same commit batch as the scripts/boot_coder_cdi.sh rootless fix (4e993f1a, already committed — see §4).

3.2 Dual-wire facade — full HTTP E2E

docs/qa/dualwire_live_e2e_20260711T130459Z/RESULTS.md (commit 8a309004) proves the complete live path `curl → gin router → wireFacadeAuthMiddleware → chatCompletions/ anthropicMessages → resolveLLMProvider("helixllm"/"local") → llm.OpenAICompatibleProvider → http://localhost:18434 (live coder) → real answer → back through the same translation layer:`

- **401** on an unauthenticated request (fail-closed, confirmed before the request ever reaches provider resolution).
- **200 + nonce echo** on both `/v1/chat/completions` (OpenAI shape) and `/v1/messages` (Anthropic shape), each with a **fresh, independently generated nonce** (HELIXCODE-FULLHTTP-E2E-203036440e6c and . . . -f6a30d57b156) echoed verbatim by the same live coder.
- **Live tool-call wire-shape divergence:** the same coder, given the same `get_weather` tool definition and a nonce-bearing prompt, was driven through both wire shapes — OpenAI's `function.arguments` renders as a **JSON-encoded string**, Anthropic's `content[].input` renders as a **JSON object** — both carrying the identical live nonce (HELIXCODE-FULLHTTP-E2E-292679c493a1), proving the divergence is a genuine translation of one underlying `llm.ToolCall`, not two independently fabricated responses.
- **Coder-untouched confirmation:** every call was a read-only `GET /v1/models` or `POST /v1/chat/completions` — no config write, no restart (§11.4.122).
- **Key-leak-free confirmation:** only a test-only, self-documented non-secret key was used; `grep` for real credential patterns over the report and raw evidence returned no hits.

3.3 Lane-B multi-instance serving (co-resident with the coder)

docs/qa/lane_b_liveproof_20260711T130533Z/RESULTS.md (commit 8244b33e, plus `benchmark_results.json` raw data and `benchmark_script.py` the exact harness executed) proves Mistral-Nemo-Instruct-2407-Q4_K_M (Lane-B, port :18435) was booted **co-resident** with the live 30B coder (port :18434) via `submodules/helix_llm/cmd/agentgen-boot`, rootless podman + NVIDIA CDI, on the shared RTX 5090:

- **Admission:** live `nvidia-smi` read immediately before boot — free 12650 MiB ≥ need 9216 MiB + 2048 MiB headroom → ADMIT-OK.
- **Co-residence:** both :18434 and :18435 answered `GET /v1/models` simultaneously post-boot; the coder's `llama-server` PID (1980342) was **identical** before and after Lane-B's full admit → boot → benchmark → teardown cycle — mechanical proof the coder was never restarted.

- **Single-stream throughput:** mean 163.67 tok/s / median 163.34 tok/s — matches the 2026-07-08 non-co-resident baseline, i.e. co-residence did not degrade Lane-B's own throughput.
- **3-parallel concurrent:** 3/3 succeeded, 2.73 s wall-clock, per-request tok/s dropping $\sim 2.2\times$ under 3-way contention (consistent with continuous-batching sharing one GPU).
- **Tool-calling/arithmetic correctness:** "What is 2+2?" $\rightarrow$ "4" via the chat-templated endpoint — a genuine methodology finding is documented in-report (§11.4.102): the raw /v1/completions continuation endpoint returned empty for a closed question, root-caused as a test-harness-endpoint mismatch (not a Lane-B defect), then fixed by switching to /v1/chat/completions.
- **Teardown:** VRAM restored to ~ 12.6 GiB free (16 MiB delta = normal driver noise), Lane-B container/port fully removed, coder PID unchanged throughout.

3.4 HelixQA — coder-bench + coder-concurrency banks, live against the coder

docs/qa/helixqa_live_vnv_20260711T130200Z/RESULTS.md (commit 930cfddb) re-ran two HelixQA banks live against the resident coder with freshly rebuilt analyzers (bin/helixqa-verify-coder-bench, bin/helixqa-verify-coder-concurrency, both built from unmodified current source immediately before use — §11.4.108 SOURCE $\rightarrow$ ARTIFACT):

- **helixllm_coder_bench.yaml:** 7/7 PASS — p50/p95/p99 latency and tok/s at graduated concurrency (1/10/50/100), TTFT streaming case (p50=3 ms, p95=4 ms, p99=4 ms, well under the 5000 ms budget), plus a golden-good and a golden-bad self-validation case.
- **helixllm_coder_concurrency.yaml:** 6/6 PASS — all-ok / no-loss / nonce-integrity / response-consistency confirmed across 50 concurrent requests (0 drops, 0 duplicate/canned responses), plus golden-good and golden-bad self-validation.
- **Self-validation (§11.4.107(10)):** both analyzers' golden-bad fixtures genuinely produced a raw pass=false (never a hardcoded/metadata-only true), correctly inverted to a case PASS only because the bank declares --expect-fail.
- **Fresh live §1.1 mutation check** (not merely cited from the bank's 2026-07-08 authoring metadata): both analyzers were rebuilt with an unconditional Pass = true line injected immediately before the ExpectFail inversion, run against their golden-bad fixtures, and shown to flip case_result from PASS to **FAIL** — proving the assertion logic is genuinely load-bearing. The source files were reverted from /tmp backups

immediately after, confirmed by an empty `git diff --stat` and no MUTATED residue (§11.4.84 working-tree quiescence held throughout).

- **Coder-untouched:** same PID (1980342) and identical `/health + /props` responses before and after the full 13-case run; every call was read-only.
- **Scope note** (stated honestly in the source report): DDoS, chaos, memory, and race dimensions for the coder are covered by separate banks not re-run in this task — those were previously live-run per `docs/qa/phase1_providers_rerun_20260708T204553Z/` and prior commits.

3.5 Generative (image/video) live-proof — PENDING

No `docs/qa/generative_liveproof_*` directory exists on disk at the time of writing (`find docs/qa -maxdepth 1 -iname "*generative*"` returns nothing, and `find docs/qa -maxdepth 1 -type d -newer docs/qa/helixqa_live_vnv_20260711T130200Z` also returns nothing, confirming no newer generative-evidence directory has landed since the last confirmed evidence in this fleet). Per the SDD ledger (`.superpowers/sdd/progress.md`, section “R41-F”), this capability is owned by background stream `ad9a782848e068b2a`, dispatched as a GPU-freed backfill immediately after Lane-B’s teardown released VRAM, described as:

“fast-lane GENERATIVE live-proof (FLUX-schnell/WAN fits free ~12.6GiB, coder untouched, §11.4.107 self-validated analyzer, honest-BLOCKED if HF_TOKEN-gated model absent)”

This row is marked PENDING per §11.4.6 — no metric is claimed or invented for it. When that stream lands its `RESULTS.md`, this summary should be updated (or superseded) to fill in the row.

4. §11.4.111 rootless coder-boot fix

The coder was down at the start of the 2026-07-11 session (`podman start helixllm-coder` failing). Root cause, per the SDD ledger and confirmed by reading the committed fix script (`scripts/boot_coder_cdi.sh`, commit `4e993f1a`, “*fix(§11.4.111): make coder-boot fully rootless — CDI to ~/.config/cdi + CDI_SPEC_DIRS (no root/su/sudo)*”):

- The coder container binds its GPU via the CDI name `nvidia.com/gpu=all`.
- The CDI spec was **pinned to a stale device path** (`/dev/dri/card0`) from before a host reboot re-enumerated DRM devices to `card1` — a textbook §11.4.111 “resolve by stable name, not enumeration index” violation.
- No CDI spec had been (re)generated at all (`nvidia-ctk cdi list` reported 0 devices; `/etc/cdi` and `/var/run/cdi` were absent).
- **Fix, entirely rootless** (no `su/sudo` needed): `nvidia-ctk cdi generate --output=~/.config/cdi/nvidia.yaml` regenerates a fresh, correctly-

resolved spec into the user's own XDG config dir, then
CDI_SPEC_DIRS=~/.config/cdi podman start helixllm-coder points
the rootless podman invocation at it — no /etc/cdi write, no privilege
escalation, no image rebuild, no container recreation.

- The rewritten script is a **permanent boot-guard**: it is safe to re-run on any future DRM re-enumeration.
- Per the SDD ledger, the operator's su password was offered but **not stored** by the agent (§11.4.10/CONST-042 leak-avoidance) — the rootless path made storing/using it unnecessary. The ledger further notes the operator may wish to rotate that password out of caution since it briefly appeared in a transcript; that rotation decision is the operator's, not restated as fact here beyond what the ledger records.

Result: the coder came up **rootless**, live at :18434, and its liveness was immediately confirmed by a fresh nonce echo (§3.1) before any of the other three live-proven streams began exercising it.

5. Honest boundary — what this document does NOT establish

Per §11.4.6 (no-guessing) and §11.4.123 (rock-solid-proof-or-research), this summary is explicit about its limits:

- **This document re-ran nothing.** Every metric above is transcribed from an already-committed RESULTS.md (or, for the coder-boot capability, an already-produced but not-yet-committed evidence pair). If any underlying evidence file is later found to be inaccurate, this summary inherits that inaccuracy — it is an index, not an independent re-verification.
- **Generative capability is genuinely PENDING**, not degraded-but-passing. No claim is made about FLUX/WAN2 image/video generation working or not working as of this document's creation.
- **DDoS, chaos, memory, and race dimensions of the §11.4.169 coverage enumeration were NOT re-run in this fleet** for the coder — the HelixQA report explicitly scopes itself to benchmarking/performance and concurrency/atomicity only, citing prior separate bank runs (docs/qa/phase1_providers_rerun_20260708T204553Z/ and earlier commits) for those other dimensions. This summary does not claim otherwise.
- **Operator-gated items remain open** (per the SDD ledger, outside this task's scope to resolve): constitution submodule push, root/submodule push to all upstreams (§11.4.113 ff-only, CONST-043 explicit approval), rotation of the previously-leaked Gemini API key and the HF token exposed in a transcript, merge-to-main + production tag (§11.4.167/§11.4.151), §11.4.185 manual QA-team final confirmation, and the cerebras dead-code-removal / replicate-provider-wiring decisions. None of these are addressed or resolved by this

document — it only indexes the already-committed live evidence produced so far.

- **The coder-boot evidence directory is currently untracked** (?? in git status) — it is cited here because it exists on disk and its content was verified, but it had not yet been committed by its owning stream at the time this summary was written. Its commit hash is therefore listed as “commit pending” rather than a concrete SHA.
- No secret values, API keys, or tokens are reproduced anywhere in this document (§11.4.10) — all evidence that mentions redacted credentials in the source RESULTS.md files is summarized here without those values.

6. Sources verified

- docs/qa/coder_boot_liveproof_20260711T125759Z/RESULTS.md + response.json — read directly, 2026-07-11 (this session).
- docs/qa/dualwire_live_e2e_20260711T130459Z/RESULTS.md — read directly, commit 8a309004 confirmed present via `git log/git show --name-only`.
- docs/qa/lane_b_liveproof_20260711T130533Z/RESULTS.md — read directly, commit 8244b33e confirmed present via `git log/git show --name-only`.
- docs/qa/helixqa_live_vnv_20260711T130200Z/RESULTS.md — read directly, commit 930cfddb confirmed present via `git log/git show --name-only`.
- scripts/boot_coder_cdi.sh — read directly, commit 4e993f1a confirmed present via `git log/git show --stat`.
- .superpowers/sdd/progress.md (section “R41-F”) — read directly for the narrative context (fleet composition, generative-stream status, rootless boot root-cause) attributed explicitly to the ledger throughout this document.
- `find docs/qa -maxdepth 1 -iname "*generative*"` and `find docs/qa -maxdepth 1 -type d -newer docs/qa/helixqa_live_vnv_20260711T130200Z` — both run 2026-07-11, both empty, confirming the generative-evidence directory genuinely does not yet exist (not merely unsearched).